

*SELENE design map, companion to the BRONTE Chimborazo sheet (KER-BRO-TRK-001). One machine, one ridge: **EOS-3** (Ø3.0 m bore, Ø2.6 m pod, 7 t bulk per shot). Bore fixed across all five phases; only the track grows. Phase 1b baseline: 2.9 km at 100 g, escape speed 2.38 km/s, 8.7 MW south-pole solar. Total build estimate: \$2.0B–\$2.5B (mid-case, see Panel F). Mission one: cargo and satellite dispensers launched from the Moon and returned to Earth or delivered to cislunar orbit. The site map uses real NASA LRO/LOLA laser topography (20 m/px polar DEM); track alignment is conceptual. All physics re-derived June 2026.*

ANNEX A

SELENE, South Pole Design Map

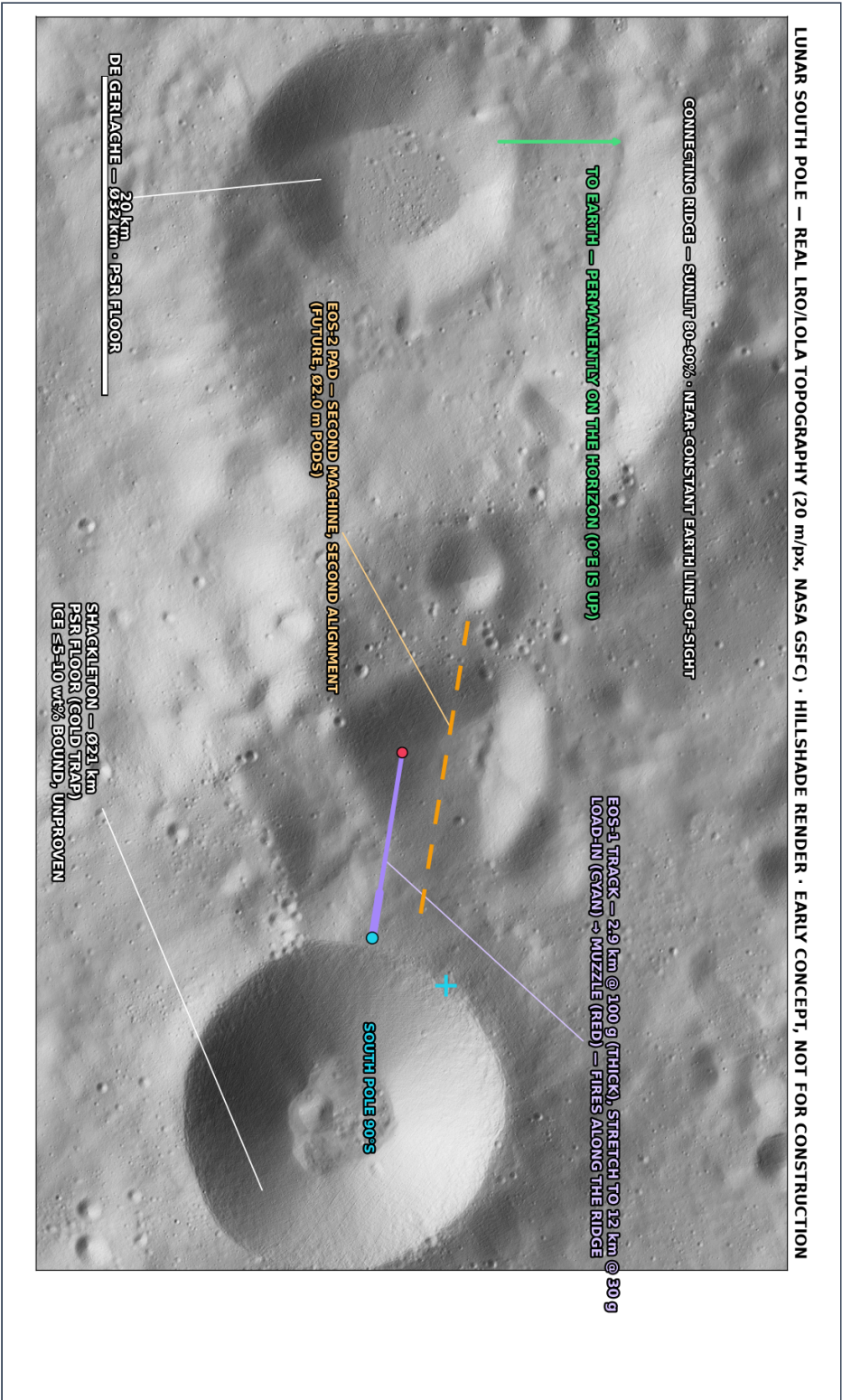
KER-SEL-TRK-002 • REV D • EARLY-VERSION TEST MOCK-UP • NOT FOR CONSTRUCTION

WHAT IS INSIDE

- ▶ Panel A, Site: real LRO/LOLA laser topography, Shackleton / Connecting Ridge ~89.7°S
- ▶ Panel A2, Where on the Moon · what does not exist here · two machines (EOS-1 / EOS-2)
- ▶ Panel B, EOS-1 open-track section: no tube, the Moon is the vacuum vessel
- ▶ Panel C, Side profile: the raised track on its regolith embankment (true 1:1)
- ▶ Charts, track-length vs g, Δv destinations, ridge illumination, engineering

Each large panel is on its own page, rotated to read at full size, turn the page (or your screen) a quarter-turn clockwise.

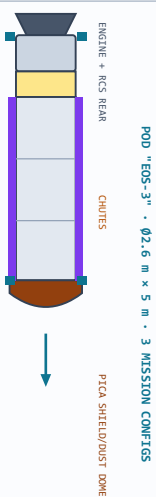
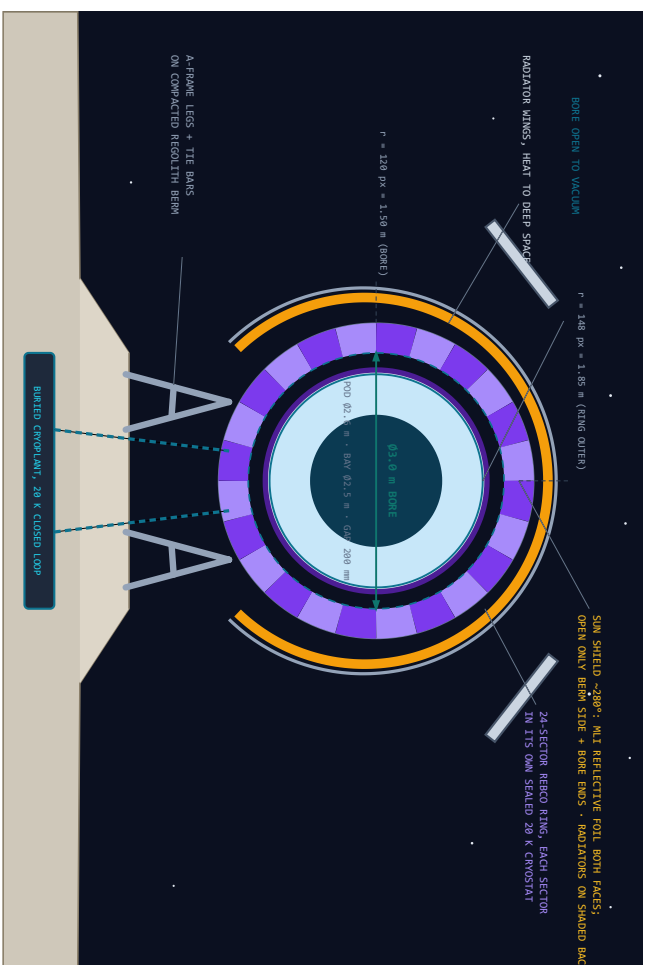
PANEL A, SITE: REAL LRO/LOLA TOPOGRAPHY, SHACKLETON / CONNECTING RIDGE, ~89.7°S



PANEL B, EOS-3 OPEN-TRACK SECTION, NO TUBE: THE MOON IS THE VACUUM VESSEL - Ø3.0 m BORE

EOS-3 SECTION · Ø3.0 m BORE · POD Ø2.6 m · GAP 200 mm · PAYLOAD BAY Ø2.5 m

SCALE: ~80 px/m • 24-SECTOR REBCO RING @ 20 K IN SEALED CRYOSTATS • WRAP-AROUND SUN SHIELD ~280° • RING OUTER Ø3.70 m



CONFIG A – BULK CARGO:

PELLETS (REGOLITH/ICE/METAL), 7,069 kg @ 500 kg/m³ IN 2 m POD

CONFIG B - PACKAGED CARGO:

3 SWAP RACKS • PRESSURE VESSELS • MANUFACTURED GOODS

~3,000-5,000 kg USEFUL LOAD • 1

CONFIG C - SATELLITE DISPENSER:

SAT STACK IN BAY 1-3 • DISPENSING

ONBOARD KICK Δv 200-400 m/s $\rightarrow$ Cislunar Orbit, L1/L2, GSO, LEO

ALL 3 CONFIGS: SAME RING, SAME TRACK, SAME DEPARTURE SPEED
HARDENED TO 100% - ARTILLERY ELECTRONICS: 1E 000-20 000 000 000

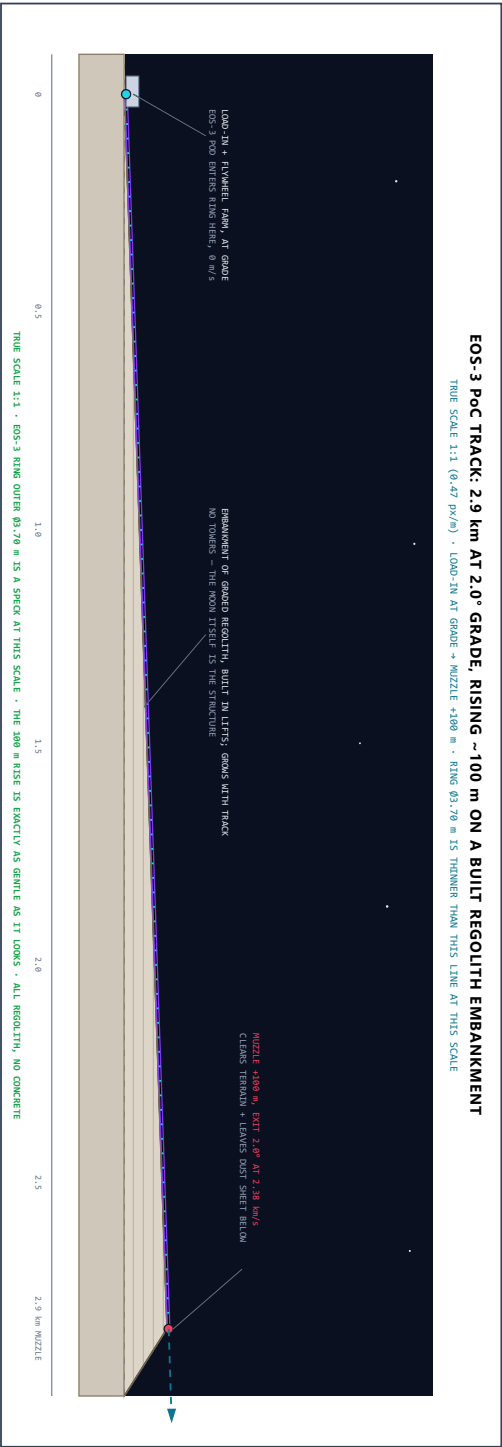
PICA HEAT SHIELD: JUST DOME AT LAUNCH. ENTRY SHIELD AT EARTH
 MAINTAINED 1000 G. AN ILLUMINATED ELECTRONICS. 15,000-20,000 G. QUALITY

1. **THE STATE OF TEXAS, COUNTY OF DALLAS, ss. I, _____, Clerk of the County Court, do hereby certify that the within and foregoing is a true and correct copy of the original as the same appears from the records of the County Court of Dallas County, Texas.**

THE HONEST HARD PARTS (LUNAR EDITION, Ø3 m)

1. DUST. CLIM LOFTED AT PPM RELEASE; ELECTROSTATIC ON RIDGE.
 2. DUST. CLIM LOFTED AT PPM RELEASE; ELECTROSTATIC ON RIDGE.
 3. THE PLUSE .079 MM IN Z 2-4 = 2.37 CM. FLAMEL FROM CHARGED SLOWLY BY 8.7 MW SOLAR. DAMPED FAST. MASS LOSS FIRST.
 4. CATCHING. MISSION ONE: EARLY AMPHIBIOUS IS THE MITT (SHIELD + CHUTES). CUSTOMER REDUNDANCY IS THE MAIN PROBLEM.
 5. ICE INSTABILITY. SHACKLETON 55-40 WEG. UNCONTINUED. RECOGNIZ + MEAL CLOSE THE EARLY BODINESS WITHOUT ICE.
- MISSING FROM THIS LIST IS THE POINT: NO TUBES, NO MUDROOM, NO HEATING, NO WATER, NO RESERVE VARIATIONS, NO FUEL.

PANEL C, SIDE PROFILE: THE RAISED TRACK ON ITS EMBANKMENT (TRUE SCALE 1:1)



PANEL A2, WHERE ON THE MOON • WHAT DOES NOT EXIST • EOS-3 MACHINE SPECS

WHERE ON THE MOON?



HERE, SOUTH POLE (89.7°S)

PHOTO: REAL NEARSIDE • MAP: REAL LRO/LOLA LASER DATA
SHACKLETON RIM + CONNECTING RIDGE: 88-92% ANNUAL SUN

WHAT DOES NOT EXIST ON THIS SHEET

NO VACUUM TUBE — THE TRACK IS ALREADY IN HARD VACUUM
NO PLASMA WINDOW — NO ATMOSPHERE TO KEEP OUT
NO AERO-SHELL — NO NEEDLE NOSE — NO AIR TO CUT THROUGH
NO ISOLATION GATES — NO BLAST SHUTTERS — NO TUBE BREACH
NO SONIC BOOM — NO LAUNCH HEATING — NO DRAG, EVER

EVERY HARD SUBSYSTEM ON THE BRONTE SHEET IS DELETED BY ONE SENTENCE: THE MOON HAS NO ATMOSPHERE.
WHAT REMAINS IS THE RING, THE POWER, AND THE COLD.

WHAT REPLACES THEM: DUST + 250-390 K THERMAL SWINGS + GW-CLASS PULSE STORAGE + CATCHING THE CARGO AT THE OTHER END.

SITE SERVICES: RIDGE SOLAR 80-92% UPTIME • PSR COLD SINK NEXT DOOR • REGOLITH ~250 K AT 2 m DEPTH, NATURAL CRYO BUFFER

MAP DATA: NASA LRO/LOLA POLAR DEM, 20 m/px (LDEM_875S_20M), POLAR STEREO, RENDERED AS HILLSHADE. A REAL SURVEY.

CHOSEN BORE: Ø3 m. SAME RING ACROSS ALL 5 PHASES.
TRACK GRONIS; BORE DOES NOT CHANGE.

EOS-3 • Ø3 m BORE • THE CHOSEN MACHINE

PHASE 1b PoC: 2.9 km • POD Ø2.6 m • BORE Ø3.0 m • GAP 200 mm
7,069 kg BULK PER SHOT (500 kg/m³, 2 m POD) • 3,000-5,000 kg CARGO
ENERGY 0.79 MWh/SHOT • PEAK ~2.37 GW • RECHARGE ~5 min @ 8.7 MW
~100 SHOTS/HR BULK OR ~10 LARGE CARGO SHOTS/HR

THROUGHPUT: ~72,000 t/yr AT 8.7 MW (POWER-LIMITED, NOT BORE-LIMITED)
THROUGHPUT RULE: 8,378 t/MWh/yr AT ESCAPE SPEED
SAME TONNAGE ALL BORE SIZES: BORE BUYS SHOT MASS, NOT ANNUAL MASS

REBCO: 2,809 km TAPE • 6.7 t • \$281M @\$100/m • \$13M DELIVERY
24-SECTOR RING @ 20 K, 0.6 m PITCH • 4,833 RINGS OVER 2.9 km
ALL 5 PHASES USE IDENTICAL RING: TRACK EXTENDS, RING STAYS

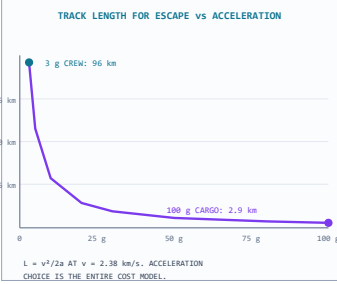
SCALING LADDER: 2.9 km → 12 km → 30 km → 50 km → 96 km
ESCAPE (2.38) • CARGO + (2.66) • HIGH-SPD (4.20) • LLO/3g • ESC/3g

FIRST BILL IS A ROCKET BILL: 6.7 t REBCO + ELECTRONICS + SOLAR PLANT
CO-MANIFESTED ON 2-3 STARSHIP LANDER FLIGHTS • ~\$254M DELIVERY

RETURN MISSIONS: CARGO, SAMPLES, SATELLITE DISPENSERS (SEE PANEL E)
POD'S PICA HEAT SHIELD DOUBLES AS DUST DOME AT LAUNCH AND ENTRY SHIELD FOR DIRECT ATMOSPHERIC RECOVERY AT EARTH.
BUILD ESTIMATE: \$2.00-\$2.50 MID-CASE (SEE PANEL F)

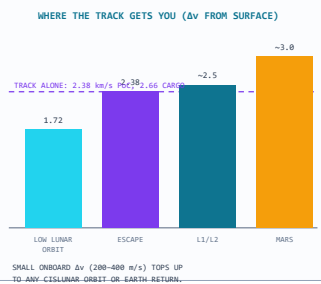
PANEL D, ENGINEERING CHARTS: TRACK LENGTH • DESTINATIONS • ILLUMINATION • ENERGY

TRACK LENGTH FOR ESCAPE vs ACCELERATION



L = $v^2/2a$ at $v = 2.38$ km/s. ACCELERATION CHOICE IS THE ENTIRE COST MODEL.

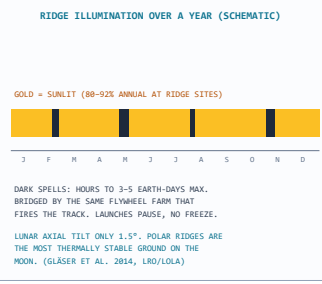
WHERE THE TRACK GETS YOU (Δv FROM SURFACE)



TRACK ALONE: 2.38 km/s POD 382.66 CARGO

SMALL ONBOARD Δv (200-400 m/s) TIPS UP TO ANY CISELUNAR ORBIT OR EARTH RETURN.

RIDGE ILLUMINATION OVER A YEAR (SCHEMATIC)

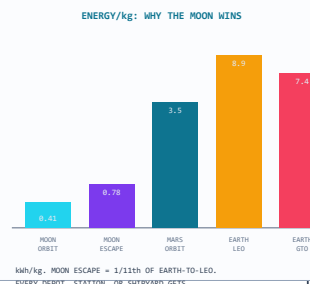


GOLD = SUNLIT (80-92% ANNUAL AT RIDGE SITES)

DARK SPELLS: HOURS TO 3-5 EARTH-DAYS MAX.
BRIDGED BY THE SAME FLYWHEEL FARM THAT FIRES THE TRACK. LAUNCHES PAUSE, NO FREEZE.

LUNAR AXIAL TILT ONLY 1.5°. POLAR RIDGES ARE THE MOST THERMALLY STABLE GROUND ON THE MOON. (GLÄSER ET AL. 2014, LRO/LOLA)

ENERGY/kg: WHY THE MOON WINS



MWh/kg. MOON ESCAPE = 1/11th OF EARTH-TO-LEO.
EVERY DEPUT, STATION, OR SHIPPING GETS CHEAPER MASS FROM THE MOON THAN FROM EARTH.

PANEL E, RETURN MISSIONS: FROM MOON TO EARTH AND CISELUNAR SPACE

MISSION TYPE 1: DIRECT EARTH ENTRY
SAMPLES, HIGH-VALUE CARGO (He-3, ZBLAN, CRYSTALS)

3-DAY TRANSIT



PHYSICS:

- LAUNCH AT $v = 2.38$ km/s (ESCAPE FROM MOON)
- FREE-COAST 3 DAYS TO EARTH VICINITY
- EARTH GRAVITY ACCELERATES TO ~11 km/s ENTRY
- PICA HEAT SHIELD (APOLLO-CLASS REENTRY)
- CHUTES DEPLOY AT ~10 km ALTITUDE
- OCEAN OR LAND RECOVERY

BEST FOR: SAMPLES, He-3 (IF WINDING VIABLE), ZBLAN FIBER, PHARMA CRYSTALS, SMALL RETURNS
NOTE: POD REACHES ~11 km/s AT EARTH, SAME AS APOLLO CH.

MISSION TYPE 2: SATELLITE DISPENSER
SMALLSATS, RELAY BIRDS, LUNAR-MANUFACTURED PAYLOADS



PHYSICS:

- POD CARRIES SAT STACK (3 BAYS, Ø2.5 m × ~4 m EACH)
- SATS HARDENED TO 100 g (MILITARY ELECTRONICS: 20,000 g)
- POD COAST 3 DAYS → RELEASES SAT STACK
- SAT ONBOARD Δv 200-400 m/s → TARGETS FINAL ORBIT
- L1/L2 LISSAJOUS: 200 m/s. GEO TRANSFER: ~800 m/s
- LEO INSERT: ~3,100 m/s (NEEDS LARGER KICK STAGE)

BEST FOR: LUNAR RELAY SATS, L1/L2 HALO HUBS, GEO COMMS, CISELUNAR LOGISTICS NODES
KEY: THE MOON BECOMES A SATELLITE LAUNCH PLATFORM AT \$4/kg.

MISSION TYPE 3: CISELUNAR DEPOT SUPPLY
LOX/LH2, WATER, REGOLITH DISPENSERS, BULK METAL



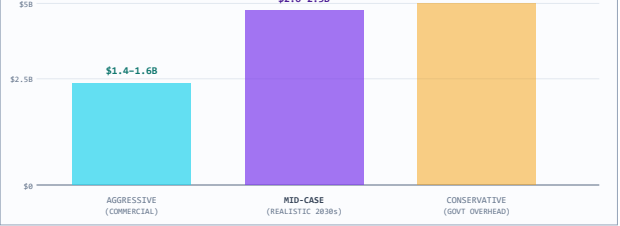
PHYSICS:

- LAUNCH BULK (ICE/LOX/H₂) AT 2.38 km/s FROM MOON
- SMALL COURSE-CORRECTION BURN (150-250 m/s)
- TARGETS THE DEPOT DIRECTLY
- POD DOCKS OR DEPOT CATCHES WITH BERTHING ARM
- PROPELLANT TRANSFER, POD RETURNED TO MOON

BEST FOR: LOX SUPPLY (~\$4/kg), WATER, RADIATION SHIELDING PANELS, STRUCTURAL METAL INGOTS

PANEL F, BUILD COST SUMMARY (REVISED ESTIMATE, Ø3 m, PHASE 1b)

TOTAL BUILD COST ESTIMATE — Ø3 m BORE, PHASE 1b
FIGURES 10-20% BELOW UNREVISED ESTIMATES. ALL FIGURES ±50%.



AGGRESSIVE (COMMERCIAL) MID-CASE (REALISTIC 2030s) CONSERVATIVE (GOVT OVERHEAD)

COST CATEGORY BREAKDOWN (MID-CASE, Ø3 m)

FIXED (ALL BORE SIZES)

- R&D + DESIGN + EARTH TEST \$780M
- SOLAR ARRAYS (8.7 MW, 43.5 t) MFG+DELIVERY \$100M
- CRYOGENIC PLANT (20 K, 15 t) \$15M
- CONSTRUCTION ROBOTS (20 t) \$460M
- POWER STORAGE (FLYWHEELS/CAPS, 30 t) \$135M
- ELECTRONICS + COMMS + CONTROL \$215M
- OPERATIONS (10-YR RUNWAY) \$215M

VARIABLE (Ø3 m BORE)

- REBCO TAPE (2,809 km @ \$100/m) \$281M
- REBCO DELIVERY (6.7 t @ \$2,000/kg) \$13M
- TRACK STRUCTURE, GUIDES, ALIGNMENT \$110M
- POD/BUCKET FLEET (10 UNITS) \$100M

TOTAL MID-CASE ~\$2.52B
COST / kg (10 yr × 72,000 t/yr = 720 Mt): ~\$3.50/kg vs STARSHIP \$200-2,000/kg

ALL FIGURES ±50%. REVISION FROM PRIOR ESTIMATE: 10-20% REDUCTION (ISRU CREDIT + COMMERCIAL MODEL + VOLUME REBCO PRICING)